

Problem F. Competição

Time Limit 1000 ms

Mem Limit 262144 kB

Tavas is a cheerleader in the new sports competition named "Pashmaks".



This competition consists of two part: swimming and then running. People will immediately start running R meters after they finished swimming exactly S meters. A winner is a such person that nobody else finishes running before him/her (there may be more than one winner).

Before the match starts, Tavas knows that there are n competitors registered for the match. Also, he knows that i -th person's swimming speed is S_i meters per second and his/her running speed is r_i meters per second. Unfortunately, he doesn't know the values of R and S , but he knows that they are real numbers greater than 0.

As a cheerleader, Tavas wants to know who to cheer up. So, he wants to know all people that might win. We consider a competitor might win if and only if there are some values of R and S such that with these values, (s)he will be a winner.

Tavas isn't really familiar with programming, so he asked you to help him.

Input

The first line of input contains a single integer n ($1 \leq n \leq 2 \times 10^5$).

The next n lines contain the details of competitors. i -th line contains two integers S_i and r_i ($1 \leq S_i, r_i \leq 10^4$).

Output

In the first and the only line of output, print a sequence of numbers of possible winners in increasing order.

Examples

Input	Output
3 1 3 2 2 3 1	1 2 3

Input	Output
3 1 2 1 1 2 1	1 3